Module Project 3

Literature Review & Simulation Notes

ECE 6580 — Temperature Transducers **(Option A: Research)**

Built from the assignment prompt and five journal papers (2018–2024), plus an equation-based simulation attempt.

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1 Overview

Temperature measurement beyond the classical triad of RTDs, thermocouples, and thermistors has become increasingly relevant as systems demand sensing in environments where wiring is impractical, electromagnetic interference is severe, or the sensor must survive conditions that destroy metallic junctions. This report surveys five such techniques drawn from recent literature, covering the underlying physics of each, what the instrument actually measures, the reported performance, and the practical limitations that matter when choosing between them. The FBG technique is then taken further with an equation-based simulation that connects the paper's reported sensitivity and spectral width to achievable temperature resolution as a function of interrogator noise.

2 Sensing Technique Overview

Table 1 maps each technique from its temperature-dependent physical mechanism to what the instrument actually measures, and identifies the dominant practical error source. The five techniques are discussed in detail in the sections that follow.

Table 1. Lookup map: how temperature is transduced, what is measured, and what limits accuracy in each technique.

Technique	Temperature dependence	Measured variable	Practical accuracy limit
Dielectric resonator	Permittivity shifts EM resonance frequency	RF resonance frequency	Reader frequency drift; resonance broadening from losses; environmental permittivity loading
SAW (wireless)	Elastic constants change acoustic phase velocity	RF echo spectrum / time delay	Electrode degradation at high T; thermal cycling drift; RF link SNR and EMI
FBG	Thermo-optic effect + thermal expansion shift Bragg condition	Optical wavelength (peak location)	Strain cross-sensitivity; interrogator wavelength resolution; peak noise
Luminescent FIR	Boltzmann population ratio between thermally coupled states	Emission intensity ratio	Optical path stability; shot noise; material aging; spectral background
MRI PRF shift	Water proton resonance frequency shifts with T	MR phase (temperature map)	Motion artifacts; susceptibility changes; scanner drift; phase reference stability

3 Literature Review

The techniques reviewed here share an important characteristic: they all produce a frequency, wavelength, phase, or intensity ratio as their primary output rather than a raw voltage or resistance. This makes them naturally compatible with digital readout chains and tends to suppress certain classes of drift that plague analog sensors. The five techniques span microwave, acoustic, optical, and imaging domains, which also means their practical tradeoffs are quite different from one another — the right choice depends heavily on the deployment environment, the required temperature range, and what infrastructure is already available.

3.1 Microwave Dielectric Resonator Sensing (frequency shift)

A dielectric resonator confines electromagnetic energy in a mode whose resonance frequency is set by the cavity geometry and the material's effective permittivity ε_{eff}. Because permittivity is temperature-dependent, the resonance frequency shifts predictably with temperature,

$$f_r(T) \propto \frac{1}{\sqrt{\varepsilon_{\text{eff}}(T)}}, \tag{1}$$

meaning the resonator itself serves as both the field-confining structure and the temperature-sensitive element — no separate transduction layer is required. In practice, a microwave reader sweeps frequency across the resonance and tracks the peak location in the S-parameters, from which temperature is inferred via a calibration curve.

Wang et al. [1] demonstrated a total frequency shift of 672 MHz over 26.7 °C to 141.6 °C, giving a sensitivity of 5.91 MHz/°C. At that sensitivity, a peak-location uncertainty of ±100 kHz translates to only ±0.017 °C in principle. In practice, however, the dominant error terms are reader reference drift, resonance broadening from dielectric losses, and parasitic loading from nearby fixtures or humidity — all of which shift f_r through mechanisms entirely unrelated to temperature. Compared to RTDs and thermistors, the sensing element requires no DC bias current, and compared to thermocouples there are no low-level thermoelectric voltages to manage, though RF interrogation hardware replaces simple signal conditioning. The approach is most compelling in sealed vessels, rotating assemblies, or embedded structures where running wires and protecting junctions are the dominant engineering challenges.

3.2 Wireless Surface Acoustic Wave (SAW) Sensing at Extreme Temperatures

A SAW device converts an RF excitation into an acoustic wave that propagates along the surface of a piezoelectric substrate and is then reconverted to an electrical signal. Because the acoustic phase velocity $v(T)$ depends on the elastic constants and density of the substrate — both of which are temperature-dependent — a characteristic frequency shifts as

$$f(T) = \frac{v(T)}{\lambda_a}, \tag{2}$$

where λ_a is the acoustic wavelength fixed by the interdigital transducer geometry. The key practical advantage is that the sensor can be interrogated wirelessly: an RF burst is transmitted to the device, and the reflected spectral response carries the temperature-encoded frequency shift back to the reader, with no electrical connection to the hot zone required.

Zhang et al. [2] demonstrated this principle at extreme temperatures using a langasite (LGS) substrate with a multilayer Pt–Rh/Zr composite electrode system specifically engineered to survive repeated excursions to 1200 °C. The reported accuracy is within 1.59 % of full scale, and the device maintains a quality factor of approximately $Q \approx 3526$ at room temperature. The engineering challenge at high temperature is electrode integrity: thermal cycling drives creep, diffusion, and recrystallization in the metallization, all of which cause the frequency–temperature calibration to drift over time. Thermocouples can reach comparable temperatures, but they require cold-junction compensation and are prone to wiring failures in rotating or inaccessible assemblies — exactly the conditions where wireless SAW sensing is most valuable. The main considerations for a SAW deployment are the RF link budget and EMI environment, since signal-to-noise at the reader directly limits temperature resolution.

3.3 Fiber Bragg Grating (FBG) Optical Wavelength Sensing

An FBG is a periodic modulation of the refractive index written into an optical fiber core, which acts as a narrow-band mirror for the Bragg wavelength $\lambda_B = 2n_{\text{eff}}\Lambda$, where n_{eff} is the effective refractive index and Λ is the grating period. Temperature changes both quantities simultaneously: the thermo-optic effect shifts n_{eff}, and thermal expansion shifts Λ, so the reflected peak moves in wavelength. Over moderate temperature ranges this shift is well approximated as linear,

$$\lambda_B(T) = \lambda_0 + S(T - T_0), \tag{3}$$

and measurement reduces to locating the peak of the reflected spectrum with an optical interrogator. An optical interrogator measures the reflected spectrum; temperature is then inferred from the peak location λ_B via the calibrated sensitivity S.

Macheso and Zekriti [3] report a sensitivity of 0.61 nm/°C and a reflection peak FWHM of approximately 7.893 nm. The FWHM matters practically because it sets a bound on how precisely the peak can be located for a given signal-to-noise ratio: narrower peaks allow more accurate centroid estimation. For IoT deployments, the interrogator architecture — spectrometer, tunable laser, or edge-filter demodulation — dominates system cost and sampling rate, and often matters more than the grating itself. The standard cross-sensitivity concern is strain: axial strain also shifts λ_B, following

$$\frac{\Delta\lambda_B}{\lambda_B} \approx (\alpha + \xi)\,\Delta T + (1 - p_e)\,\varepsilon, \tag{4}$$

where α is thermal expansion, ξ the thermo-optic coefficient, p_e the effective photoelastic constant, and ε axial strain. Temperature-only applications therefore benefit from strain-isolated packaging or a second reference grating. In exchange for this extra design consideration, FBGs offer immunity to electromagnetic interference, easy wavelength-division multiplexing of many sensors on a single fiber, and compatibility with long-distance remote sensing.

3.4 Luminescent Thermometry via Fluorescence Intensity Ratio (FIR)

Luminescent thermometry exploits the temperature dependence of radiative transitions in rare-earth-doped materials. When two excited states are thermally coupled — separated by energy ΔE — their populations obey a Boltzmann distribution, so the ratio of their emission intensities varies with temperature as

$$\mathrm{FIR}(T) = \frac{I_1(T)}{I_2(T)} \approx A \exp\left(-\frac{\Delta E}{k_B T}\right). \tag{5}$$

The power of the fluorescence intensity ratio (FIR) approach lies in measuring this *ratio* rather than an absolute intensity: excitation power fluctuations, detector gain drift, and optical coupling variations cancel out to first order, giving a readout that is inherently self-referencing.

Wei et al. [4] report that the FIR changes by as much as 90-fold between room temperature and 513 K in their Pr^{3+}-doped $(Ba_{0.7}Sr_{0.3})TiO_3$ material, with a relative sensitivity fit of $S_r(T) = 4275.1/T^2$, where the relative sensitivity is defined as

$$S_r(T) = \frac{1}{\mathrm{FIR}} \frac{d\,\mathrm{FIR}}{dT}. \tag{6}$$

The $1/T^2$ dependence means sensitivity is highest at low temperatures and decreases as the material warms, which informs the useful operating range for a given application. Practical deployment raises questions about long-term material stability under sustained heat exposure, whether the optical collection path is stable enough for the ratio assumption to hold, and how well ambient background fluorescence can be rejected. Despite these considerations, the technique is genuinely attractive for remote, contact-free measurement where electrical connections and metallic sensors are undesirable — biological systems, microfluidics, and high-voltage environments being natural examples.

3.5 MRI Proton Resonance Frequency (PRF) Shift Thermometry

In MRI, the phase of the acquired signal depends on the local Larmor frequency, which is sensitive to both the static magnetic field and the chemical environment of the protons being imaged. Water protons experience a temperature-dependent chemical shift because the hydrogen-bond network changes with temperature, altering the electron shielding of each proton. This effect is approximately linear over the

clinically relevant range, and the resulting temperature change can be estimated from the measured phase difference between a baseline and a subsequent acquisition:

$$\Delta T \approx \frac{\Delta\phi}{\gamma B_0 \alpha \, \mathrm{TE}}, \tag{7}$$

where γ is the gyromagnetic ratio, B_0 the static field strength, α the PRF thermal coefficient, and TE the echo time. The output is not a point measurement but a *temperature map* — a spatial image of temperature change across a tissue volume — which is what makes PRF thermometry uniquely suited to monitoring thermal therapies such as MR-guided high-intensity focused ultrasound (HIFU) and laser-induced thermal therapy, where knowing the spatial extent of heating is as important as knowing the peak temperature.

Blackwell et al. [5] provide a thorough clinical review of how these principles are applied in practice. The main limitations they identify are MRI-specific rather than fundamental to thermometry. Motion is the most serious: respiration, cardiac pulsation, and patient movement all corrupt the phase reference and produce apparent temperature errors that are difficult to distinguish from real heating. Susceptibility changes at tissue interfaces also shift the local resonance frequency independently of temperature. Clinical practice therefore relies on motion compensation strategies, careful sequence design, and reference subtraction schemes to maintain thermometry accuracy during interventions. Compared to any of the other techniques reviewed here, PRF thermometry requires by far the most complex and expensive infrastructure, but it is the only method that provides real-time, non-invasive, volumetric temperature monitoring inside the human body.

3.6 Synthesis: How do these Techniques Compare?

These five papers collectively show us how modern temperature sensing has moved beyond the classical triad of RTDs, thermocouples, and thermistors. The underlying principle of the alternatives discussed here are that they all produce a naturally digital or ratio-metric output, which includes a frequency, a wavelength, a phase, or an intensity ratio. As a result, rather than a raw voltage or resistance that requires careful analog signal conditioning. This tends to reduce certain classes of drift and makes long-distance or wireless transmission more practical. In addition, Table 2 summarizes the key differences.

Table 2. High-level comparison of the five sensing techniques.

Technique	Physical mechanism	Readout	Key system tradeoffs
Dielectric resonator	$\varepsilon(T)$ shifts EM mode frequency	RF resonance frequency	Passive and wireless-capable; needs RF hardware; cross-sensitive to humidity and nearby objects
SAW	$v(T)$ changes acoustic delay	RF echo spectrum	Wireless and passive; survives harsh environments; electrode longevity and RF link are the challenges
FBG	Thermo-optic + expansion shift λ_B	Optical wavelength	EMI immune; easily multiplexed; strain cross-sensitivity requires isolation or referencing
Luminescent FIR	Boltzmann population ratio	Emission intensity ratio	Self-referencing ratio metric; contact-free; needs optical excitation and stable collection path
MRI PRF shift	Proton chemical shift with T	MR phase (temperature map)	Volumetric non-invasive mapping; requires MRI scanner; motion and susceptibility are key errors

4 Simulation attempt (equation-based) of the FBG paper

The Macheso et al [3] was selected for simulation since its core model is explicit and equation-driven: a linear wavelength shift with a clearly stated sensitivity and a characterized spectral peak width. This makes an equation-based simulation both faithful to the paper and practically illuminating, of which the simulation can directly show how interrogator noise translates into temperature uncertainty, which is the central engineering design question for any FBG deployment.

4.1 Approach and implementation

The simulation is built on the linear Bragg wavelength model of Eq. (3), using $\lambda_0 = 1550\,\text{nm}$, $T_0 = 25\,^\circ\text{C}$, $S = 0.61\,\text{nm}/^\circ\text{C}$, and $\text{FWHM} = 7.893\,\text{nm}$ as reported. To simulate a wavelength-domain measurement, the reflected spectrum at each temperature is modeled as a Gaussian peak centered at $\lambda_B(T)$:

$$R(\lambda;T) = \exp\left(-\frac{(\lambda-\lambda_B(T))^2}{2\sigma^2}\right), \quad \sigma = \frac{\text{FWHM}}{2\sqrt{2\ln 2}}. \tag{8}$$

The simulation was implemented in Python using NumPy for numerical computation and Matplotlib for visualization. For this we implemented four experiments. First, the Bragg wavelength was swept across a temperature range of $25\,^\circ\text{C}$ to $100\,^\circ\text{C}$ to verify the linear sensitivity and produce a calibration curve. Second, the full reflected spectrum was synthesized at six representative temperatures to visualize how the peak migrates across the wavelength axis. Third, additive Gaussian noise was superimposed on the spectrum at $T = 60\,^\circ\text{C}$ and temperature was estimated by argmax over 5000 independent noise realizations, producing an empirical distribution of peak-location errors. Fourth, the noise level was swept

logarithmically and the resulting temperature standard deviation $\sigma_T = \sigma_\lambda / S$ was recorded at each level to map out the interrogator noise–resolution tradeoff and the full code is shown in Appendix A.

4.2 Results and Interpretation

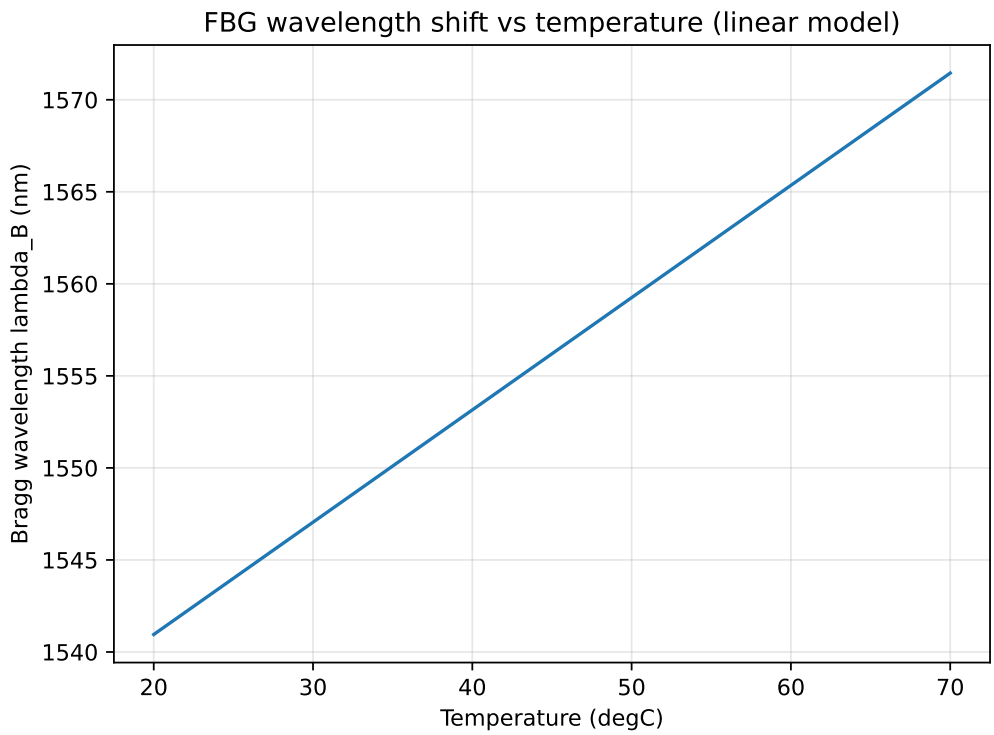

Figure 1. FBG Bragg wavelength shift versus temperature using the linear model (Eq. (3)).

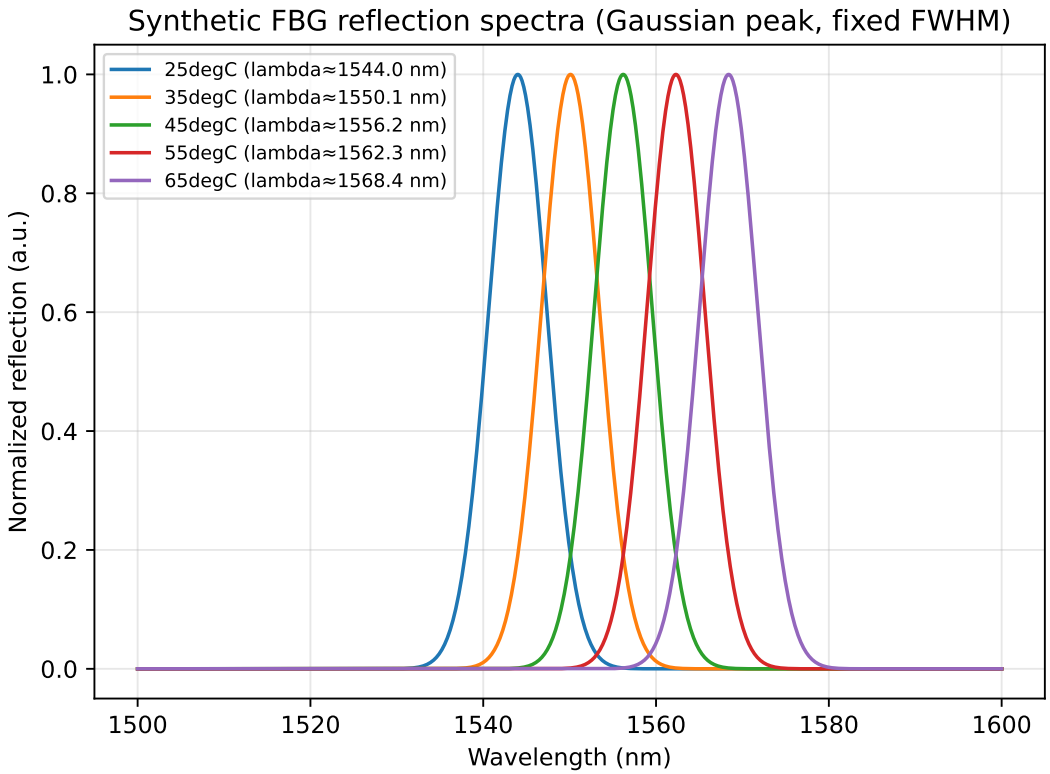

Figure 2. Synthetic reflection spectra (Gaussian peaks) at several temperatures, illustrating how λ_B moves with T.

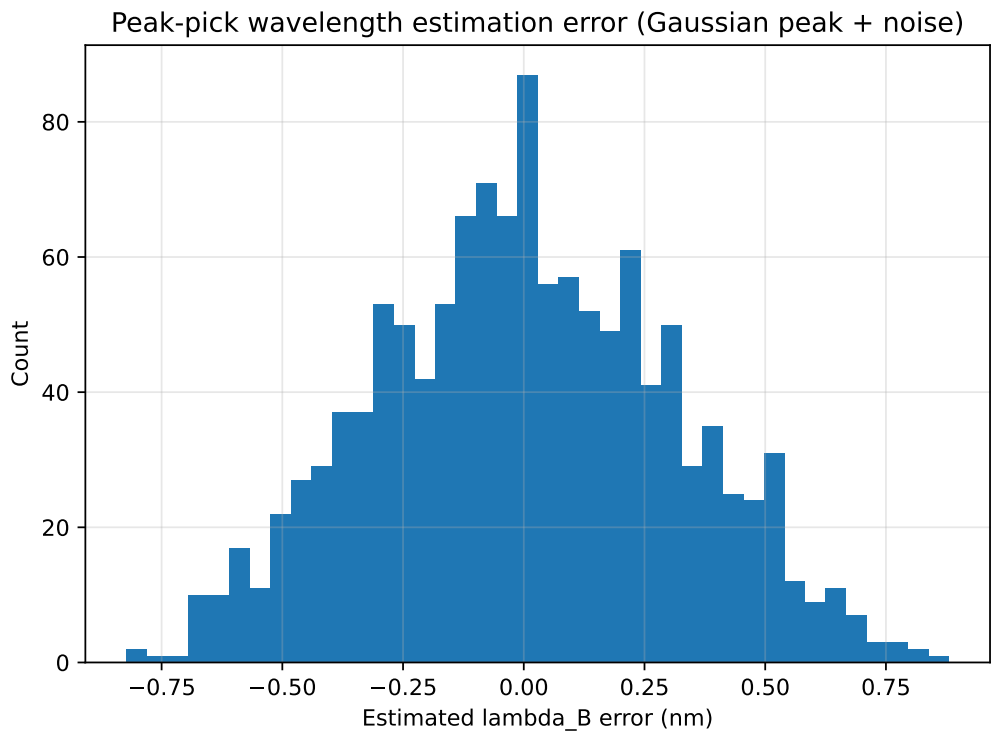

Figure 3. Peak-picking wavelength estimation error under additive noise (simple estimator).

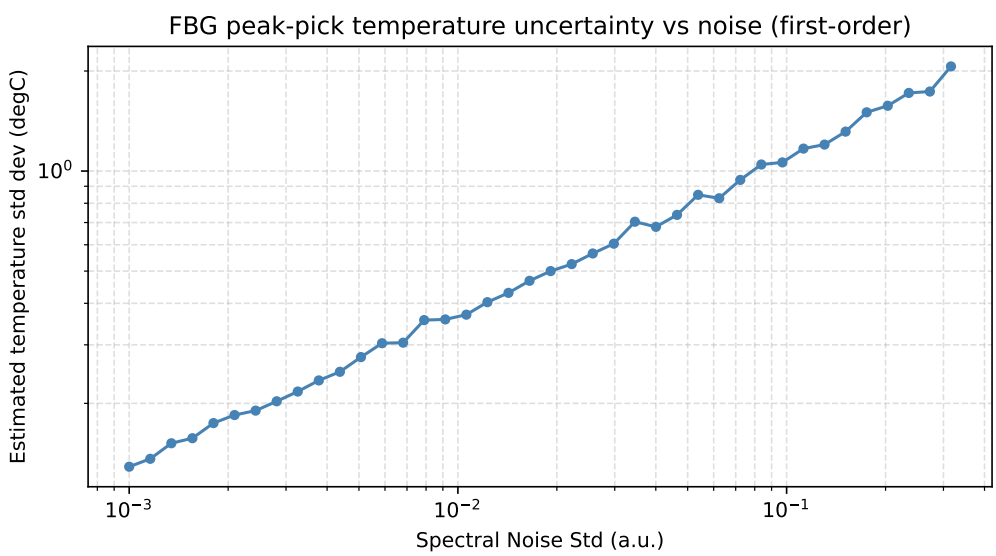

Figure 4. Estimated temperature uncertainty σ_T versus spectral noise level, mapped through sensitivity S.

Figure 1 confirms the expected linear relationship between temperature and Bragg wavelength. Figure 2 shows how the reflection peak migrates across the wavelength axis as temperature rises, with the spacing between adjacent peaks directly proportional to S. The noise trials in Figures 3 and 4 reveal the core

engineering tradeoff: temperature resolution is set by $\sigma_T \approx \sigma_\lambda / S$, so with $S = 0.61\,\mathrm{nm}/^\circ\mathrm{C}$, achieving $\sigma_T = 0.1\,^\circ\mathrm{C}$ requires the interrogator to locate the spectral peak to within about 0.06 nm. Whether that is achievable in practice depends entirely on the interrogator architecture and its noise floor, which is a grating spectrometer and a tunable-laser system differ by orders of magnitude in wavelength resolution.

The model is deliberately simple, which is appropriate given that the paper itself is primarily analytical rather than experimental. A higher-fidelity simulation would add strain cross-sensitivity, temperature-dependent linewidth, wavelength-dependent source noise, non-Gaussian lineshapes arising from grating apodization, and systematic calibration errors in the interrogator. Each of these would widen the error distributions in Figures 3 and 4, making the idealized curves shown here a best-case bound rather than a realistic prediction.

A Python Code for Reproducibility

The following Python script will produce the figures that are included in the simulation section.

```python
"""
FBG Temperature Sensor Simulation  --  ECE 6580 Module Project 3
Simulates FBG response based on Macheso & Zekriti (2024), IJIN vol. 5, pp.
    224-230.
"""
import numpy as np
import matplotlib.pyplot as plt
import os

# Parameters from the paper
lambda0 = 1550.0        # centre Bragg wavelength at T0 [nm]
T0      = 25.0          # reference temperature [degC]
S       = 0.61          # sensitivity [nm/degC]
FWHM    = 7.893         # reported FWHM [nm]
sigma   = FWHM / (2 * np.sqrt(2 * np.log(2)))

os.makedirs("figs", exist_ok=True)

# 1. Bragg wavelength vs temperature
T_arr    = np.linspace(25, 100, 300)
lambda_B = lambda0 + S * (T_arr - T0)

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(T_arr, lambda_B, color='steelblue', linewidth=2)
ax.set_xlabel("Temperature (degC)")
ax.set_ylabel("Bragg Wavelength (nm)")
ax.set_title("FBG Bragg Wavelength vs Temperature")
ax.grid(True, linestyle='--', alpha=0.5)
fig.savefig("figs/fbg_lambda_vs_T.pdf", dpi=150)
plt.close(fig)

# 2. Reflection spectra at multiple temperatures
lambda_ax = np.linspace(1530, 1620, 2000)
T_list = [25, 40, 55, 70, 85, 100]
colors = plt.cm.plasma(np.linspace(0.15, 0.85, len(T_list)))
```

```python

fig, ax = plt.subplots(figsize=(8, 4.5))
for T, c in zip(T_list, colors):
    lB = lambda0 + S * (T - T0)
    R  = np.exp(-0.5 * ((lambda_ax - lB) / sigma) ** 2)
    ax.plot(lambda_ax, R, color=c, label=f"{T} degC")
ax.set_xlabel("Wavelength (nm)")
ax.set_ylabel("Normalised Reflectance")
ax.set_title("FBG Reflection Spectra at Several Temperatures")
ax.legend(title="Temperature", fontsize=9, loc="upper left")
fig.savefig("figs/fbg_spectra_multiT.pdf", dpi=150)
plt.close(fig)

# 3. Peak estimation error histogram
rng         = np.random.default_rng(42)
noise_level = 0.02
N_trials    = 5000
T_test      = 60.0
lB_true     = lambda0 + S * (T_test - T0)
R_clean     = np.exp(-0.5 * ((lambda_ax - lB_true) / sigma) ** 2)

peak_errors = []
for _ in range(N_trials):
    R_noisy = R_clean + rng.normal(0, noise_level, size=len(lambda_ax))
    lB_est  = lambda_ax[np.argmax(R_noisy)]
    peak_errors.append(lB_est - lB_true)

fig, ax = plt.subplots(figsize=(7, 4))
ax.hist(peak_errors, bins=60, color='steelblue', edgecolor='white', alpha=0.85)
ax.axvline(0, color='crimson', linewidth=1.5, linestyle='--', label='True peak')
ax.set_xlabel("Peak Estimation Error (nm)")
ax.set_ylabel("Count")
ax.legend()
fig.savefig("figs/fbg_peak_est_error_hist.pdf", dpi=150)
plt.close(fig)

# 4. Temperature uncertainty vs noise level
noise_levels = np.logspace(-3, -0.5, 40)
sigma_T_list = []
for nl in noise_levels:
    errs = [(lambda_ax[np.argmax(R_clean + rng.normal(0, nl, size=len(lambda_ax)))]
            - lB_true) / S  for _ in range(500)]
    sigma_T_list.append(np.std(errs))

fig, ax = plt.subplots(figsize=(7, 4))
ax.loglog(noise_levels, sigma_T_list, 'o-', color='steelblue', markersize=4)
ax.set_xlabel("Spectral Noise Std (a.u.)")
ax.set_ylabel("Temperature Uncertainty (degC)")
ax.set_title("Estimated sigma_T vs Spectral Noise Level")
ax.grid(True, which='both', linestyle='--', alpha=0.4)
fig.savefig("figs/fbg_temp_sigma_vs_noise.pdf", dpi=150)
plt.close(fig)

```

```
print("All figures saved to figs/")
```

References

[1] Q. Wang, Y. Liu, M. Zhang, X. Bai, and Z. Gao. A simple dielectric resonator-based sensor for temperature measurement. *Results in Physics*, 27:104481, 2021.

[2] H. Zhang, D. Mu, Z. Zhang, J. Zhang, J. Sun, and H. Jin. Surface acoustic wave sensors for wireless temperature measurements above 1200 degree celsius. *Sensors*, 24(15):4945, 2024.

[3] P. S. Macheso and M. Zekriti. Modelling and analysis of fiber bragg grating temperature sensor for Internet of Things applications (FBG-4-IoT). *International Journal of Intelligent Networks*, 5:224–230, 2024.

[4] T. Wei, Z. Wu, Y. Zhao, Q. Chen, and S. Liu. Novel optical temperature sensor based on emission in Pr^{3+} doped ferroelectric $(Ba_{0.7}Sr_{0.3})TiO_3$. *RSC Advances*, 8(42):23471–23476, 2018.

[5] J. Blackwell, M. J. Kraśny, A. O'Brien, K. Ashkan, J. Galligan, M. Destrade, and N. Colgan. Proton resonance frequency shift thermometry: A review of modern clinical practices. *Journal of Magnetic Resonance Imaging*, 55(2):389–403, 2022.